
MATH550 | HOMEWORK #0

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Abstract

This report assumes familiarity with the assignment as provided by Dr. Sprinkle. Most problems only include a plot and brief description as requested.

1 Problem 1: Finite Differences & Problem 2: Second-order Derivatives

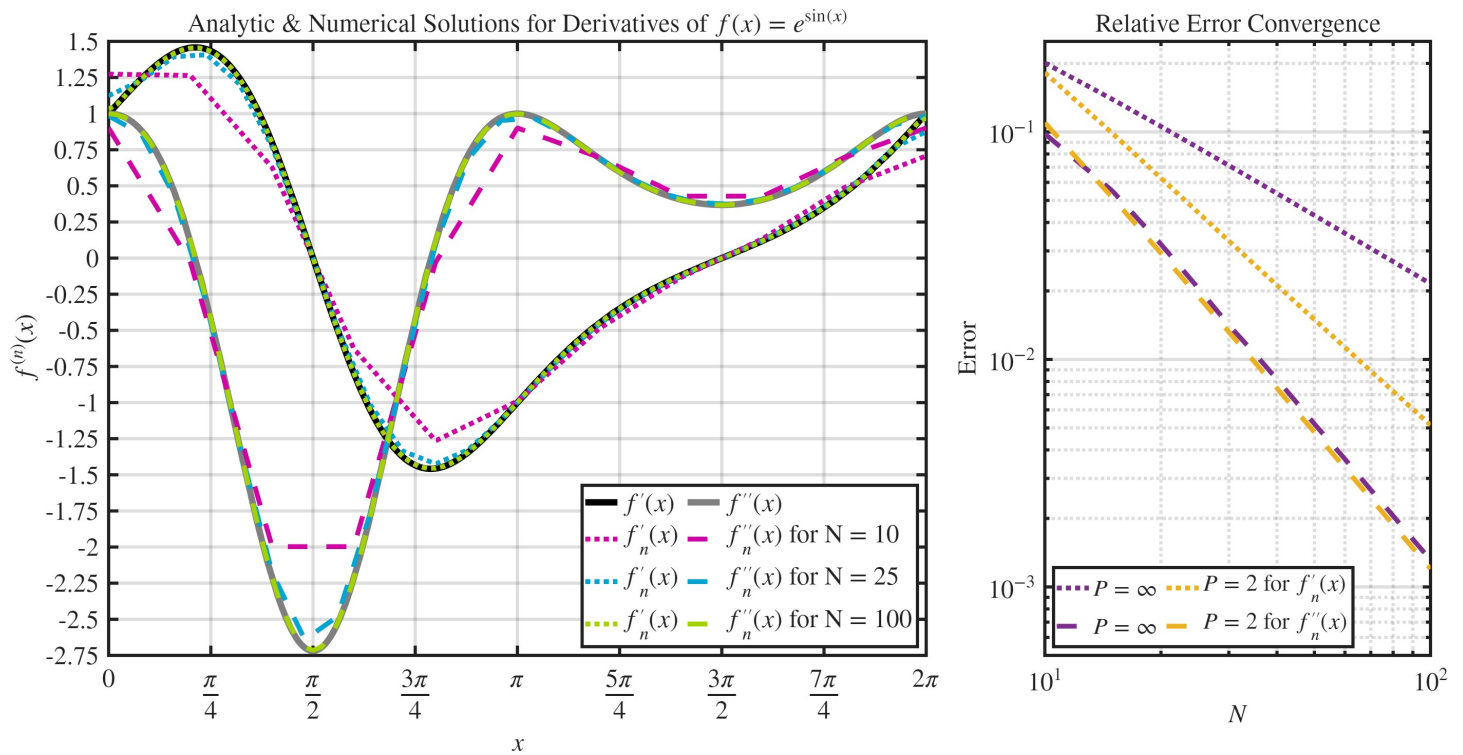


Figure 1: Function and Error Plot

The $P = 2$ error plot slope is $-3/2$ for the first derivative because there is a mix of first order (used on the boundaries) and second order (used for the non-boundaries) accurate methods.

2 Problem 3: ODE

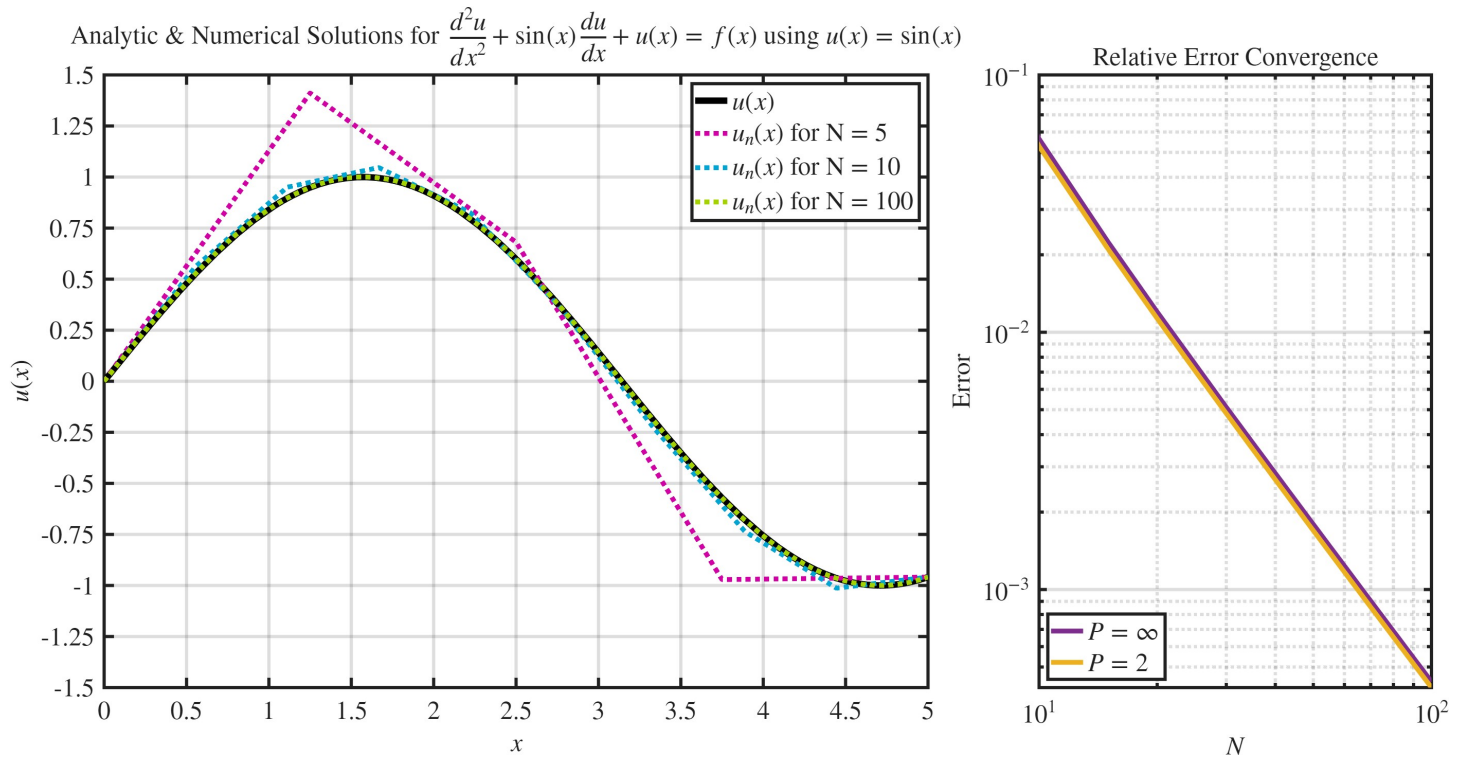


Figure 2: Function and Error Plot

For the solution $u(x) = \sin(x)$ to be valid, the forcing function must be set to $f(x) = \sin(x) \cos(x)$. The code sets the boundaries as follows: $u(0) = \sin(0) = 0$ and $u(5) = \sin(5) \approx -0.95892$. The slope of the convergence plot is -2 because second-order finite difference methods are employed over the entire domain.

3 Problem 4: Poisson Equation

Analytic and Finite Difference Solutions to Poisson Equation $\nabla^2 u = f(x, y)$ with the Test Solution $u(x, y) = \sin(x) \cos(y)$

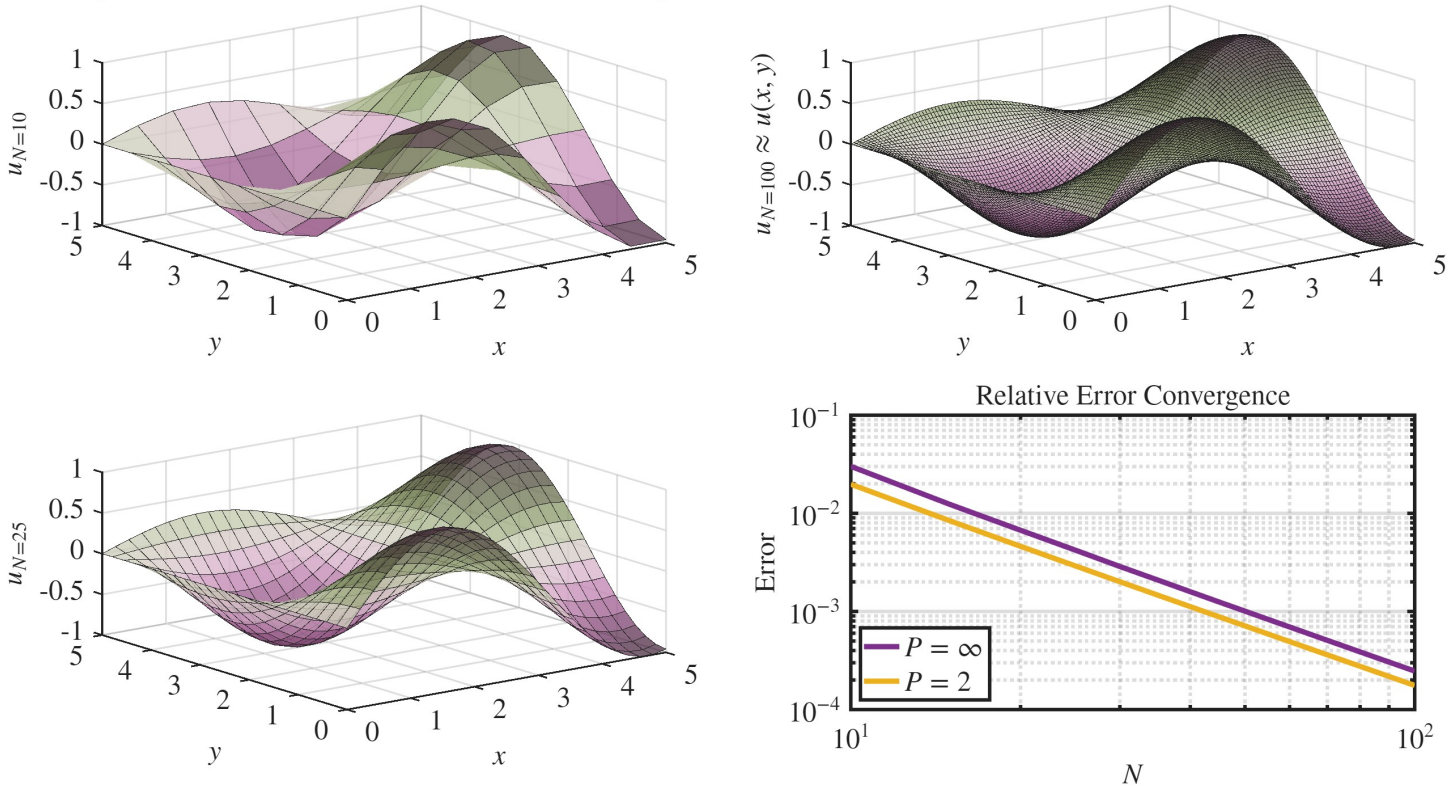


Figure 3: Function and Error Plot

3.1 Part 1

For the solution $u(x) = \sin(x) \cos(y)$ to be valid, the forcing function must be set to $f(x) = -2\sin(x) \cos(y)$. To determine the boundary conditions, $u(x) = \sin(x) \cos(y)$ is evaluated for each edge of the domain. The slope of the convergence plot is -2 because second-order finite difference methods are employed over the entire domain.

3.2 Part 2

If Neumann (derivative) boundary conditions were to be employed the PDE would be indeterminate and have infinity many solutions. This is because there would be no way to set the "constant of integration" for the function. To fix this, in addition to Neumann conditions being applied, any singular point on the grid could be set to any value which would "pin" the solution to the PDE in place. Alternatively, an equation could be added that forces the sum of all grid values to be zero. This would effectively "center" the solution to the PDE around 0.

4 Commented Code

Commented MATLAB Code is provided below.

4.1 Problems 1 & 2

```

clc, clear, close all
window_size_for_export = [0, 50, 1920, 975];

colors = {'#d300a5', '#00a5d3', '#a5d300'}; %triadic colors
background_color = [1,1,1]; %white
analytic_color = 'k'; %black
font = 24; %fontsize
lw = 4.5; %linewidth
lwlrg = 6; %linewidth large
lwsml = 3; %linewidth small

%% Problem Parameters

domain = [0, 2*pi]; % interval
f = @(x) exp(sin(x)); % funtion
df_dx = @(x) cos(x).*exp(sin(x)); % funtion 1st
    derivative (analytic)
d2f_dx2 = @(x) -sin(x).*exp(sin(x)) + cos(x).^2.*exp(sin(x)); % funtion 2nd
    derivative (analytic)

% Jack's Parameters
N_range = [10, 100];
N_plot = [10, 25, 100]; % choose 3 values of N to show on the plot (must be in
    N_range and divisible by 5)

%% Discretization and Numerical Solution

N = linspace(N_range(1), N_range(2), (N_range(2)-N_range(1))/5+1); % grid sizes
h = (domain(2)-domain(1))./N; % step sizes
x = cell(length(N),1); % setup x
    scale for each N
f_primes_n = cell(length(N),2); % setup
    results (first column is first derivative, 2nd is 2nd)

for j = 1:length(N)
    x{j} = linspace(domain(1), domain(2), N(j)+1); % discretize the domain for
        each N

```

```

    [f_primes_n{j,1}, f_primes_n{j,2}] = deal(zeros(N(j)+1,1)); % setup cell
        arrays to store results for f'(x) and f''(x)
end

for j = 1:length(N)
    f_primes_n{j,1}(1) = (f(x{j}(1)+h(j)) - f(x{j}(1) - h(j))) / h(j); % forward
        difference
    f_primes_n{j,1}(end) = (f(x{j}(end)) - f(x{j}(end)-h(j))) / h(j); %
        backward difference
end

for j = 1:length(N)
    for i = 2:length(f_primes_n{j,1})-1
        f_primes_n{j,1}(i) = (f(x{j}(i)+h(j)) - f(x{j}(i)-h(j))) / (2*h(j)); %
            central difference (1st derivative)
    end

    for i = 1:length(f_primes_n{j,2}) %looping through entire array since using
        wrap around indexing
        %technically I'm not wrapping around because I'm lazy but this will
        %still work and yield the same answer
        f_primes_n{j,2}(i) = (f(x{j}(i)+h(j)) - 2*f(x{j}(i)) + f(x{j}(i)-h(j)))
            / h(j)^2; % central difference (2nd derivative)
    end
end

%% Part 1 Plot

fig = figure(1);
set(gcf, 'Position', window_size_for_export);

t = tiledlayout(1,20);
nexttile([1, 13]);
fplot(df_dx,domain,"Color",analytic_color,LineWidth=lwlr)
hold on
fplot(d2f_dx2,domain,"Color",[.5,.5,.5],LineWidth=lwlr)
for i = 1:length(N_plot)
    index = find(N == N_plot(i));
    plot(x{index},f_primes_n{index,1},"Color",colors{i},LineWidth=lw,LineStyle
        =":")
end

```

```

    plot(x{index},f_primes_n{index,2},"Color",colors{i},LineWidth=lw,LineStyle
        ="--")
end
legend('NumColumns', 2, 'Orientation', 'horizontal');
lgd = legend('$f^{\prime}(x)$','$f^{\prime\prime}(x)$','$f_n^{\prime}(x)$' ...
            ,strcat('$f_n^{\prime}\prime(x)$ for N = ',
                num2str(N_plot(1))) ...
            ,'$f_n^{\prime}(x)$' ...
            ,strcat('$f_n^{\prime}\prime(x)$ for N = ',
                num2str(N_plot(2))) ...
            ,'$f_n^{\prime}(x)$' ...
            ,strcat('$f_n^{\prime}\prime(x)$ for N = ',
                num2str(N_plot(3))) ...
            , 'Interpreter', 'latex',
            Location='southeast');

xlabel('$x$', 'Interpreter', 'latex');
ylabel('$f^{\prime(n)}(x)$', 'Interpreter', 'latex');
title('Analytic & Numerical Solutions for Derivatives of $f(x) = e^{\sin(x)}$'
    , 'Interpreter', 'latex');
ylim([-2.75 1.5])
yticks(-2.75:0.25:1.5)
ax1 = gca;
ax1.LineWidth = lwsm1;
ax1.XTick = 0 : pi/4 : 2*pi;
ax1.TickLabelInterpreter = 'latex';
ax1.XTickLabel = {'$0$', '$\frac{\pi}{4}$', '$\frac{\pi}{2}$', '$\frac{3\pi}{4}$', '$\pi$', '$\frac{5\pi}{4}$', '$\frac{3\pi}{2}$', '$\frac{7\pi}{4}$', '$2\pi$'};
fontsize(font,"points")
grid on
box on
set(gca, 'Color', background_color)
lgd.Color = background_color;
hText = findall(gcf, 'Type', 'text');
set(hText, 'FontWeight', 'bold');
set(gca, 'FontWeight', 'bold');

```

```
%% Part 2
```

```
[Err_p_2{1},Err_p_inf{1},Err_p_2{2},Err_p_inf{2}] = deal(zeros(length(N),1)); %
    setup arrays to hold errors

for j = 1:length(N)
    top_diff{1} = df_dx(x{j})' - f_primes_n{j,1}; % top of error equation
    Err_p_2{1}(j) = norm(top_diff{1})/norm(df_dx(x{j})); %P-2 Norm Error
    Err_p_inf{1}(j) = max(abs(top_diff{1}))/max(abs(df_dx(x{j}))); %P-inf Norm
        Error

    top_diff{2} = d2f_dx2(x{j})' - f_primes_n{j,2}; % top of error equation
    Err_p_2{2}(j) = norm(top_diff{2})/norm(d2f_dx2(x{j})); %P-2 Norm Error
    Err_p_inf{2}(j) = max(abs(top_diff{2}))/max(abs(d2f_dx2(x{j}))); %P-inf
        Norm Error
end

nexttile([1, 7]);
loglog(N,Err_p_inf{1},LineWidth=lw,Color='#7E2F8E',LineStyle=':')
hold on
loglog(N,Err_p_2{1},LineWidth=lw,Color='#EDB120',LineStyle=':')
loglog(N,Err_p_inf{2},LineWidth=lw,Color='#7E2F8E',LineStyle='--')
loglog(N,Err_p_2{2},LineWidth=lw,Color='#EDB120',LineStyle='--')

xlabel('$N$', 'Interpreter', 'latex');
ylabel('Error', 'Interpreter', 'latex');
title('Relative Error Convergence', 'Interpreter', 'latex');
fontsize(font,"points")
grid on
box on
set(gca, 'TickLabelInterpreter', 'latex')
legend('AutoUpdate', 'off');
legend('NumColumns', 2, 'Orientation', 'horizontal');
lgd2 = legend('$P=\infty$', '$P=2$ for $f_n^{\prime}(x)$', '$P=\infty$', '$P=2$
    for $f_n^{\prime\prime}(x)$', 'Interpreter', 'latex', Location='southwest');
ax2 = gca;
ax2.LineWidth = lwsm1;
ylim([0.0005, 0.25])

exportgraphics(gcf,"HW0_1and2.jpg")
```

```
%was trying to get it to render in vecor form but no luck:
% drawnow;
% exportgraphics(fig, 'HW0_1and2.pdf', 'ContentType', 'vector', '
    PreserveAspectRatio', 'on');
% saveas(fig, 'my_figure.pdf');
```

4.2 Problem 3

```
clc, clear, close all

font = 24; %fontsize
lw = 4.5; %linewidth
lwlrg = 6; %linewidth large
lwsm1 = 3; %linewidth small
window_size_for_export = [0, 50, 1920, 975];
analytic_color = 'k'; %black
colors = {'#d300a5', '#00a5d3', '#a5d300'}; %triadic colors

domain = [0,5];
N_range = [5, 100];
N = linspace(N_range(1),N_range(2),(N_range(2)-N_range(1))/5+1); % grid sizes
N_plot = [5, 10, 100]; % choose 3 values of N to show on the plot (must be in
    N_range and divisible by 5)
h = (domain(2)-domain(1))./(N-1); % step sizes
    OFF BY 1 ORGINALLY!!!! NEEDED TO BE N-1
x = cell(length(N),1); % setup x
    scale for each N

[DDM, f, u, u_actual] = deal(cell(length(N),1)); % discrete derivative matrices
    and f for each N and solution matrix

for i = 1:length(N)
    DDM{i} = zeros(N(i));
    f{i} = zeros(N(i),1);
    x{i} = linspace(domain(1),domain(2),N(i));
    DDM{i}(1,1) = 1;
    DDM{i}(end,end) = cos(domain(end));
    u_actual{i} = sin(x{i});
end

for i = 1:length(N)
```

```

    for j = 2:N(i)-1
        DDM{i}(j,j-1) = 1/h(i)^2 - sin(x{i}(j))/(2*h(i));
        DDM{i}(j,j) = 1 - 2/h(i)^2;
        DDM{i}(j,j+1) = 1/h(i)^2 + sin(x{i}(j))/(2*h(i));
    end
    for j = 1:N(i)
        f{i}(j) = sin(x{i}(j))*cos(x{i}(j));
    end
end

for i = 1:length(N)
    u{i} = DDM{i}\f{i};
end

%% Plotting
fig = figure(1);
set(gcf, 'Position', window_size_for_export);

t = tiledlayout(1,20);
nexttile([1, 13]);
hold on
fplot(@(x) sin(x),domain,"Color",analytic_color,LineWidth=lwlr)

for i = 1:length(N_plot)
    index = find(N == N_plot(i));
    plot(x{index},u{index,1},"Color",colors{i},LineWidth=lw,LineStyle=":")
end

xlabel('$x$', 'Interpreter', 'latex');
ylabel('$u(x)$', 'Interpreter', 'latex');
title('Analytic \& Numerical Solutions for $\frac{d^2 u}{dx^2} + \sin(x)\frac{du}{dx} + u(x) = f(x)$ using $u(x)=\sin(x)$', 'Interpreter', 'latex');
legend('$u(x)$' ...
    ,strcat('$u_n(x)$ for N = ',num2str(N_plot(1))) ...
    ,strcat('$u_n(x)$ for N = ',num2str(N_plot(2))) ...
    ,strcat('$u_n(x)$ for N = ',num2str(N_plot(3))) ...
    , 'Interpreter', 'latex',Location='northeast');
ylim([-1.5 1.5])
yticks(-1.5:0.25:1.5)
xticks(0:0.5:5)

```

```

ax1 = gca;
ax1.LineWidth = lwsm1;
ax1.TickLabelInterpreter = 'latex';
fontsize(font,"points")
grid on
box on

%% Error calcs

[Err_p_2,Err_p_inf] = deal(zeros(length(N),1)); % setup arrays to hold errors

for j = 1:length(N)
    top_diff = u_actual{j} - u{j}'; % top of error equation
    Err_p_2(j) = norm(top_diff)/norm(u_actual{j}); %P-2 Norm Error
    Err_p_inf(j) = max(abs(top_diff))/max(abs(u_actual{j})); %P-inf Norm Error
end

nexttile([1, 7]);
loglog(N,Err_p_inf,LineWidth=lw,Color='#7E2F8E')
hold on
loglog(N,Err_p_2,LineWidth=lw,Color='#EDB120')
legend('$P=\infty$', '$P=2$', 'Interpreter', 'latex', Location='southwest')
xlabel('$N$', 'Interpreter', 'latex');
xlim([10,100])
ylabel('Error', 'Interpreter', 'latex');
title('Relative Error Convergence', 'Interpreter', 'latex');
fontsize(font,"points")
grid on
box on
set(gca, 'TickLabelInterpreter', 'latex')
ax2 = gca;
ax2.LineWidth = lwsm1;

set(gcf, 'Position', window_size_for_export);
exportgraphics(gcf,"HW0_3.jpg")
% exportgraphics(gca, 'HW0_3.pdf', 'ContentType', 'vector');

```

4.3 Problem 4

```

clc, clear, close all

```

```

font = 24; %fontsize
lw = 4.5; %linewidth
lwlrg = 6; %linewidth large
lwsm1 = 3; %linewidth small
window_size_for_export = [0, 50, 1920, 975];

[xdomain,ydomain] = deal([0,5]);
N_range = [10, 100];
N_plot = [10, 25, 100];
N = linspace(N_range(1),N_range(2),(N_range(2)-N_range(1))/5+1); % grid sizes
h = (xdomain(2)-xdomain(1))./(N-1); % step sizes
[x,y,X,Y] = deal(cell(length(N),1)); % setup x
    scale for each N

[DDM, f, u, u_actual] = deal(cell(length(N),1)); % discrete derivative matrices
    and f for each N and solution matrix

for n = 1:length(N)
    DDM{n} = sparse(N(n)^2);
    f{n} = zeros(N(n)^2,1);
    x{n} = linspace(xdomain(1),xdomain(2),N(n));
    y{n} = linspace(ydomain(1),ydomain(2),N(n));
    [X{n},Y{n}] = meshgrid(x{n},y{n});
    u_actual{n} = zeros(length(x{n}));
    for j = 1:length(y{n})
        u_actual{n}(j,:) = sin(x{n})*cos(y{n}(j));
    end
end

for n = 1:length(N)
    DDM_row = 1;
    for j = 1:length(y{n})
        for i = 1:length(x{n})
            if i == 1 || j == 1 || i == length(x{n}) || j == length(y{n}) % on
                boundary
                DDM{n}(DDM_row, global_index(j,i,N(n))) = -2*h(n)^2;
            else
                DDM{n}(DDM_row, global_index(j,i,N(n))) = -4;
                DDM{n}(DDM_row, global_index(j,i+1,N(n))) = 1;
                DDM{n}(DDM_row, global_index(j+1,i,N(n))) = 1;
            end
        end
        DDM_row = DDM_row + length(x{n});
    end
end

```

```

        DDM{n}(DDM_row, global_index(j, i-1, N(n))) = 1;
        DDM{n}(DDM_row, global_index(j-1, i, N(n))) = 1;
    end
    f{n}(DDM_row) = -2*h(n)^2*sin(x{n}(i))*cos(y{n}(j));
    DDM_row = DDM_row + 1;
end
end
u{n} = DDM{n}\f{n};
u{n} = reshape(u{n}, N(n), N(n))';
end

function g = global_index(j, i, N)
    g = N*(j-1) + i;
end

figure(1)
i=1;
plots_out_of_order = [1,3,2];
z_label = [strcat("$u_{N}=", num2str(N_plot(1)), "}$"), strcat("$u_{N}=", num2str(
    (N_plot(2)), "}$"), strcat("$u_{N}=", num2str(N_plot(3)), "}" \approx u(x,y)
    $"$)];
for n = N_plot
    subplot(2,2,plots_out_of_order(i))
    plot_i = find(N == n);
    surf(X{plot_i}, Y{plot_i}, u{plot_i}, FaceAlpha=0.75)
    colormap(slanCM('bamo'));
    xlabel('$x$', 'Interpreter', 'latex');
    ylabel('$y$', 'Interpreter', 'latex');
    zlabel(z_label(i), 'Interpreter', 'latex');
    xticks(0:1:5)
    yticks(0:1:5)
    zticks(-1:0.5:1)
    fontsize(font, "points")
    xlim([0,5])
    ylim([0,5])
    zlim([-1,1])
    set(gca, 'TickLabelInterpreter', 'latex')
    ax1 = gca;
    ax1.LineWidth = lwsml*.5;

```

```

        i = i + 1;
end

%% Error calcs

[Err_p_2,Err_p_inf] = deal(zeros(length(N),1)); % setup arrays to hold errors

for j = 1:length(N)
    top_diff = u_actual{j} - u{j}; % top of error equation
    top_diff = top_diff(:); %flattening
    Err_p_2(j) = norm(top_diff)/norm(u_actual{j}(:)); %P-2 Norm Error
    Err_p_inf(j) = max(abs(top_diff))/max(abs(u_actual{j}(:))); %P-inf Norm
        Error
end

sgtitle('Analytic and Finite Difference Solutions to Poisson Equation  $\nabla^2$ 
    u = f(x,y) with the Test Solution  $u(x,y)=\sin(x)\cos(y)$ ', 'Interpreter',
    'latex');

subplot(2,2,4)
loglog(N,Err_p_inf,LineWidth=lw,Color='#7E2F8E')
hold on
loglog(N,Err_p_2,LineWidth=lw,Color='#EDB120')
% axis square
legend('$P=\infty$', '$P=2$', 'Interpreter', 'latex', Location='southwest')
xlabel('$N$', 'Interpreter', 'latex');
ylabel('Error', 'Interpreter', 'latex');
title('Relative Error Convergence', 'Interpreter', 'latex');
fontsize(font,"points")
grid on
box on
set(gca, 'TickLabelInterpreter', 'latex')
ax2 = gca;
ax2.LineWidth = lwsml;

set(gcf, 'Position', window_size_for_export);
exportgraphics(gcf,"HW0_4.jpg")

```

5 References

The original assignment description(s) are provided below along with scratch work.

Numerical PDEs Homework #0

For all problems asking for a graph: you do not need to write up the method you programmed. Including plots of your solution and convergence is sufficient for full credit. The ideal homework will have one, information-dense and properly labeled figure with a caption and a *brief* discussion. **You must type up your HW.** I don't care if it's word or Tex as long as it's well organized, easy to follow, and submitted as a pdf.

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.
2. Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and 3/2 respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The 3/2 slope is strange, why is that happening?

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Problem 3: ODE Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).
2. Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

$$1) \text{ backward: } \frac{\partial f}{\partial x} = \frac{f(x) - f(x-h)}{h}$$

$$\text{Central: } \frac{\partial f}{\partial x} = \frac{f(x+h) - f(x-h)}{2h}$$

$$\text{forward: } \frac{\partial f}{\partial x} = \frac{f(x+h) - f(x)}{h}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$f = e^{\sin(x)} \\ \frac{\partial f}{\partial x} = \cos(x) e^{\sin(x)}$$

(Gemma)
What I used AI for:

- 1) formatting graphs syntax
- 2) I thought 1st order would be better so I loaded case and asked how I was calculating wrong bit this or I was okay
- 3) When given AI find error (missing gbs) in p-infinity case

Problem 3: ODE Consider the ODE

$$\frac{d^2 u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

$$-\cancel{\sin(x)} + \sin(x) \cos(x) + \cancel{\sin(x)} = f(x)$$

$$f(x) = \sin(x) \cos(x)$$

BCs:

$$\sin(0) = 0$$

$$\sin(5) \approx -0.95892$$

*Solving f, BCs
converges w/
second order

$$\phi''(x) = \frac{\phi(x-\Delta x) - 2\phi(x) + \phi(x+\Delta x)}{\Delta x^2}$$

$$\frac{\phi(x+\Delta x) - \phi(x-\Delta x)}{2\Delta x} \approx \phi'(x)$$

$$\frac{d^2 u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

$$\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \sin(x) \frac{u_{i+1} - u_{i-1}}{2h} + u_i = \sin(x) \cos(x)$$

$$u_{i-1} \left(\frac{1}{h^2} - \frac{\sin(x)}{2h} \right) + u_i \left(1 - \frac{2}{h^2} \right) + u_{i+1} \left(\frac{1}{h^2} + \frac{\sin(x)}{2h} \right) = \sin(x) \cos(x)$$

$$\begin{bmatrix} 1 \\ \frac{1}{h^2} - \frac{\sin(x)}{2h} \\ 1 - \frac{2}{h^2} \\ \frac{1}{h^2} + \frac{\sin(x)}{2h} \end{bmatrix}$$

$$\cos(s) \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ \vdots \\ u_{n-1} \\ u_n \end{bmatrix} = \begin{bmatrix} \sin(x_0) \cos(x_0) \\ \vdots \\ \sin(s) \cos(s) \end{bmatrix}$$

$\uparrow$
 forces u_1 to $\sin(s)$

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).
2. Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$$

$$-\sin(x)\cos(y) - \sin(x)\cos(y) = f(x, y)$$

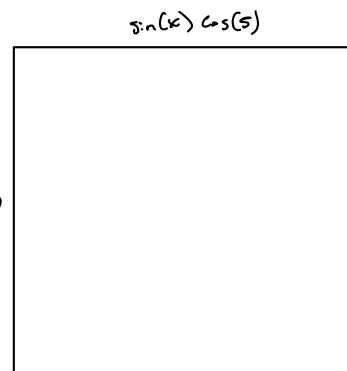
$$\frac{u_{i-1}^j - 2u_i^j + u_{i+1}^j}{h^2} + \frac{u_i^{j-1} - 2u_i^j + u_i^{j+1}}{h^2} = -2\sin(x)\cos(y)$$

$$f(x, y) = -2\sin(x)\cos(y)$$

$$\frac{u_{i-1}^j + u_{i+1}^j + u_i^{j-1} + u_i^{j+1} - 4u_i^j}{h^2} = -2\sin(x)\cos(y)$$

$$u_{i-1}^j + u_{i+1}^j + u_i^{j-1} + u_i^{j+1} - 4u_i^j = -2h^2\sin(x)\cos(y)$$

$$\begin{bmatrix} 0 & 0 & h^2 \cos(y) & 0 & 0 \end{bmatrix} \begin{bmatrix} u_i^j \end{bmatrix} = \begin{bmatrix} -2h^2 \sin(x) \cos(y) \end{bmatrix}$$



1) Why is solution not-unique

2) How do I fix it

you can fix it anywhere

fix to

now is anywhere and still satisfy PDE & BCs

Numerical PDEs Homework #0

For all problems asking for a graph: you do not need to write up the method you programmed. Including plots of your solution and convergence is sufficient for full credit. The ideal homework will have one, information-dense and properly labeled figure with a caption and a *brief* discussion. **You must type up your HW.** I don't care if it's word or Tex as long as it's well organized, easy to follow, and submitted as a pdf.

Problem 1: Finite differences Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.
2. Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and 3/2 respectively, i.e. it scales with $\frac{1}{N}, \frac{1}{N^{3/2}}$. Extra Credit: The 3/2 slope is strange, why is that happening?

Problem 2: Second-order derivatives Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Problem 3: ODE Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x)\frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Problem 4: Poisson Equation Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).
2. Modify your code to apply Neumann conditions on all boundaries. **This won't work.** Why not? – both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.